

How to Integrate Suricata with Wazuh for Network Intrusion Detection

1- Definition

Suricata is a powerful open-source intrusion detection system (IDS) and engine that analyzes network traffic (monitors network packets). **Wazuh** is a host-based IDS that monitors system files and logs.

This means Wazuh reads Suricata's alerts and displays them on the dashboard.

Host-level + network-level = visibility in one place.

Network Intrusion Detection is a digital security system for a computer network that constantly listens to and analyzes network traffic to spot suspicious activity, cyberattacks, or policy violations. If an unauthorized person tries to sneak in or a piece of malware starts acting strangely, the system immediately sounds an alarm or logs the incident so security teams can investigate.

2- Combination wazuh & Suricata notion

We can combine Wazuh and Suricata based on three features:

A- scope: Wazuh processes or checks files and logs inside the system, while Suricata checks network traffic packets and their flow. This means defense in depth.

B- Detection: Wazuh is a host-based IDS that checks for unauthorized file changes, failed logins, and suspicious processes, while Suricata checks for malware downloads, exploit attempts, port scans, and bad IPs. If we combine their power, it means full attack chain visibility.

C- Data source: Wazuh checks operating system logs, app logs, or application logs, whereas Suricata checks live traffic or pcap files. If we unify their power, it means unified host plus network events.

That combination can help us build this table

	Wazuh (Host-Based)	Suricata (Network-Based)	Combined Power
Scope	Monitors a single host	Monitors network traffic	Defense in Depth
Detects	Unauthorized file changes	Malware downloads	Full-chain attack
Data Source	Operating System Logs	Network Packets	Unified view

NB: After using Suricata, we can not only secure machines but also secure communication or network between them.

3- Configuration of suricata on wazuh Linux agent

Firstly: add the Suricata repository using this command: `sudo add-apt-repository ppa:oisf/suricata-stable`

After having this:

```
Repository: 'Types: deb
URIs: https://ppa.launchpadcontent.net/oisf/suricata-stable/ubuntu/
Suites: resolute
Components: main
'
Description:
Suricata IDS/IPS/NSM stable packages
https://suricata.io/
https://oisf.net/

Suricata IDS/IPS/NSM - Suricata is a high performance Intrusion Detection and Prevention System and Network Security Monitoring engine.

Open Source and owned by a community run non-profit foundation, the Open Information Security Foundation (OISF). Suricata is developed by
the OISF, its supporting vendors and the community.

This Engine supports:

- Multi-Threading - provides for extremely fast and flexible operation on multicore systems.
- Multi Tenancy - Per vlan/Per interface
- Uses Rust for most protocol detection/parsing
- TLS/SSL certificate matching/logging
- JA3 TLS client fingerprinting
- JA3S TLS server fingerprinting
- IEEE 802.1ad (QinQ) and IEEE 802.1Q (VLAN) support
- VXLAN support
- All JSON output/logging capability
- IDS runmode
- IPS runmode
- IDPS runmode
- NSM runmode
- eBPF/XDP
- Automatic Protocol Detection and logging - IPv4/6, TCP, UDP, ICMP, HTTP, SMTP, TLS, SSH, FTP, SMB, DNS, NFS, TFTP, KR5, DHCP, IKEv2, SNMP,
SIP, RDP
- SCADA automatic protocol detection - ENIP/DNP3/MODBUS
- File Extraction HTTP/SMTP/FTP/NFS/SMB - over 4000 file types recognized and extracted from live traffic.
- File MD5/SHA1/SHA256 matching
- Gzip Decompression
- Fast IP Matching
- Datasets matching
- Rustlang enabled protocol detection
- Lua scripting

and many more great features -
https://suricata.io/features/all-features/
More info: https://launchpad.net/~oisf/+archive/ubuntu/suricata-stable
Adding repository.
Press [ENTER] to continue or Ctrl-c to cancel.
```

press Enter.

When you will see this: `Reading package lists... Done,` it shows us that the repository has been added.

Secondly: Install Suricata as well using this command: `sudo apt install suricata`

Having this:

```
Created symlink '/etc/systemd/system/multi-user.target.wants/suricata.service' -> '/usr/lib/systemd/system/suricata.service'.
Processing triggers for libc-bin (2.43-2ubuntu2)...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.
No VM guests are running outdated hypervisor (qemu) binaries on this host
shows that Suricata has been installed.
```

Thirdly: verify the installation version using this command: `sudo suricata --version`

```
root@ubuntuuser:~# suricata --version
suricata: unrecognized option '--version'
Suricata 8.0.6 ("undefined")
```

Here you can see that Suricata version 8.0.6 has been installed.

Fourthly: Enable Suricata to start automatically on boot using this command: `sudo systemctl enable suricata`

Fifty: Download the Suricata rules using this command: `sudo suricata-update`

this

```
2/9/2026 -- 01:21:06 - <Info> -- Writing /var/lib/suricata/rules/classification.config
2/9/2026 -- 01:21:06 - <Info> -- Testing with suricata -T.
2/9/2026 -- 01:21:37 - <Info> -- Done.
```

The rules have been downloaded.

NB: Suricata will write these rules in JSON format, and then the Wazuh agent will read those JSON logs to show us attack patterns on the Wazuh dashboard.

Sixthly: Edit our configuration file. There, we will define our interface and network details (check your IP and interface first because you will need it for that).

```
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:06:24:15 brd ff:ff:ff:ff:ff:ff
    altname enx080027062415
    inet 192.168.40.222/24 brd 192.168.40.255 scope global dynamic enp0s3
        valid_lft 159084sec preferred_lft 159084sec
    inet6 2601:152:4c7e:4460:a00:27ff:fe06:2415/64 scope global dynamic noprefixroute
        valid_lft 345600sec preferred_lft 345600sec
    inet6 2601:152:4c7e:4460::32e6/128 scope global dynamic noprefixroute
        valid_lft 591082sec preferred_lft 591082sec
    inet6 2601:152:4c7e:4460::c109/128 scope global dynamic noprefixroute
        valid_lft 591085sec preferred_lft 591085sec
    inet6 fe80::a00:27ff:fe06:2415/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
```

go into the Suricata configuration file using this command: `sudo nano /etc/suricata/suricata.yaml`

Here, for the HOME_NET section, we need to define our network range.

Change this

address-groups: HOME_NET: "[192.168.0.0/16,10.0.0.0/8,172.16.0.0/12]" #HOME_NET: "[192.168.0.0/16]" #HOME_NET: "[10.0.0.0/8]" #HOME_NET: "[172.16.0.0/12]" #HOME_NET: "any"	To this	address-groups: HOME_NET: "[192.168.40.0/24,10.0.0.0/8,172.16.0.0/12]" #HOME_NET: "[192.168.0.0/16]" #HOME_NET: "[10.0.0.0/8]" #HOME_NET: "[172.16.0.0/12]" #HOME_NET: "any"
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press Ctrl+W and search for `af-packet` and enter.

Here we need to define our `interface: (enp0s3)`. After that, we have to make sure that we have these: `cluster-id is 99`, `cluster-type: cluster_flow` `defrag: yes`.

Save and exit. After that, restart your Suricata using this command: `sudo systemctl restart suricata`.

4- Configure Linux group or the Linux agent in wazuh to send Suricata logs to the Wazuh server

Do: go to our Wazuh server, go to Agent Configuration, and add this Suricata log block code:

```
<!-- Enable Suircata Logs -->  
<localfile>  
  <log_format>json</log_format>  
  <location>/var/log/suricata/eve.json</location>  
</localfile>
```

NB: To do that you have to click on Agent Management Groups (We did this on the Linux group or the Linux agent).

After that, click on Save.

NB: This code block will basically send Suricata logs to the Wazuh server. Now, we will restart our Wazuh server.

5- Trigger an IDS on wazuh agent (our Linux machine)

Do: Go to your Wazuh agent and trigger an IDS by using this command: `curl http://testmyids.com`

```
tims@wazuh:~$ curl http://testmyids.com  
uid=0(root) gid=0(root) groups=0(root)
```

Check if logs are being generated or not by type this command:

```
sudo tail -f /var/log/suricata/eve.json
```

Then your logs will start to generate.

Navigate to the **Threat Hunting** module on the Wazuh dashboard. Set the time filter to the **last 15 minutes** to review the newly generated Suricata alerts. Each entry displays a specific threat signature and description—such as potentially bad traffic—along with protocol details like HTTP and the designated log source location (eve.json).

